ECE 8920 Final Project: Multi-depot Vehicle Routing Problem with Per-vehicle Weights

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Background

- A logistics application needs to minimize the cost of a routing problem
- Some # of employees (drivers) need to visit some # of houses (customers)
- Each employee:
  - Takes off from their own house (depot)
  - Has different hourly pay, MPG, etc.
    - Different costs per driver, per graph edge traversed by said driver
- This is a modified multi-depot vehicle routing problem (MDVRP)
  - Generalization of the traveling-salesman problem (TSP) – **NP-Hard**!

Easy Problem – MDVRP

- Use WA state graph from Homework 3 as an example
  - 3 drivers, two in Tacoma, one in Pullman
  - Bellingham cannot be accessed via a cycle, so remove it
- Dantzig-Fulkerson-Johnson mixed-integer LP formulation of TSP is extensible to MDVRP [1]
- Given $G = (V, E)$ and $D \subseteq V$, let:
  - $x_{i,j}$:= Binary variable, whether or not edge $i \to j$ was traversed ($i, j \in V$)
  - $c_{i,j}$:= Weight of edge $i \to j$
  - d_i := # of vehicles at depot $i \in D$
  - $i \sim j$:= vertex i is adjacent to vertex j

Easy Problem – MDVRP

$$z = \min \sum_{i,j \in V} c_{i,j} x_{i,j}$$

$$\forall i \in V/D \quad \sum_{j \sim i} x_{i,j} = 1$$

$$\forall j \in V/D \quad \sum_{i \sim j} x_{i,j} = 1$$

$$\forall i \in D \quad \sum_{j \sim i} x_{i,j} = d_i$$

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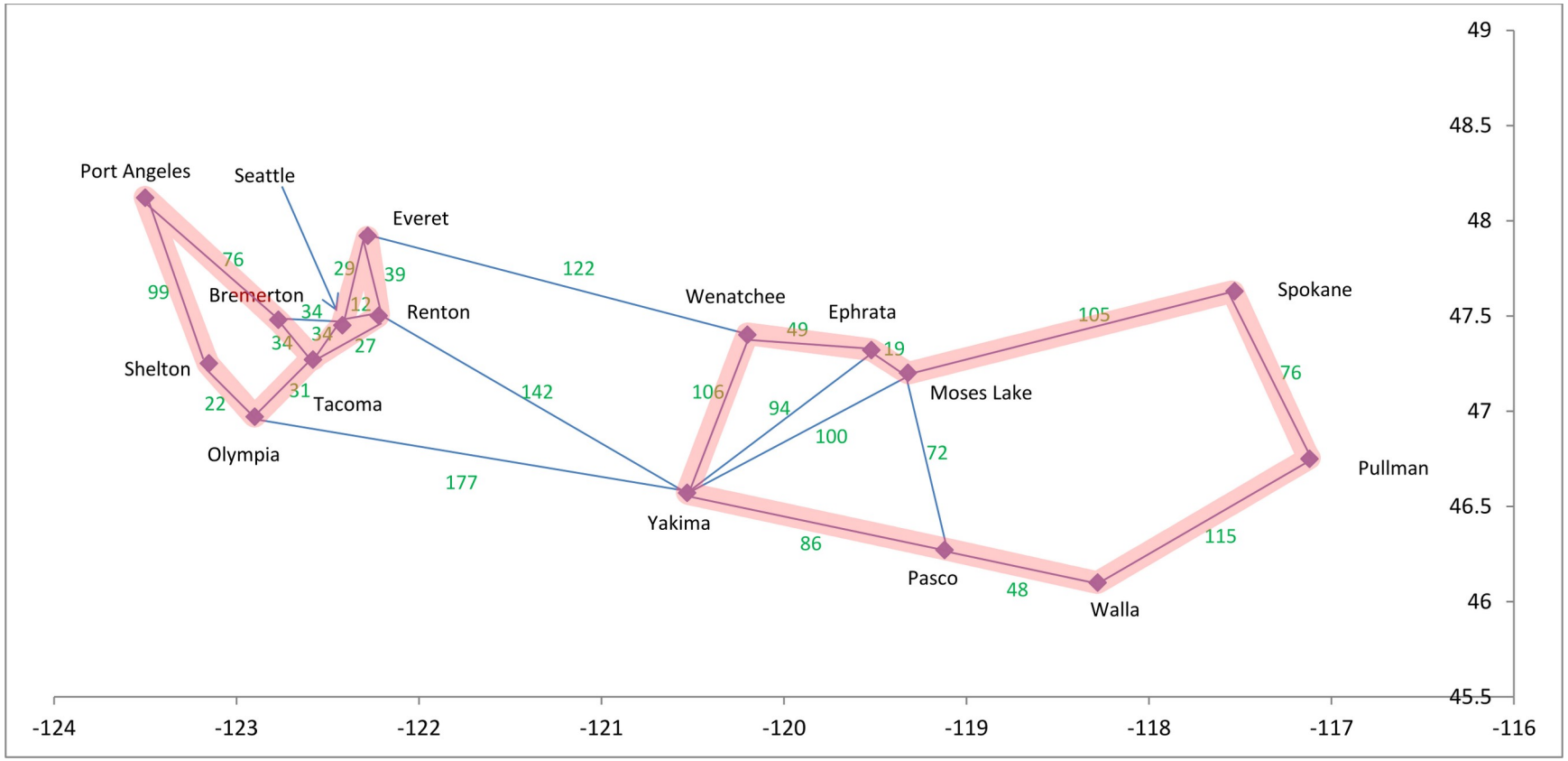
$$\forall S \subseteq V/D \text{ s.t } S \neq \emptyset \quad \sum_{j \in S} \sum_{i \notin S, i \sim j} x_{i,j} \geq 1$$

Notice: this is over the power set of V / D!

Easy Problem – MDVRP



Harder Problem – MDVRP w/ per-vehicle weights

- Previously, all drivers have the same cost for all edges
- Introduce the following biases:
  - Driver 1 from Tacoma: multiply edge weight by ½ if the destination vertex is *north* of Tacoma
  - Driver 2 from Tacoma: multiply edge weight by ½ if the destination vertex is *south* of Tacoma
  - Driver 3 from Pullman: don’t go to vertices west of Moses Lake (add M >> 0 to edges that do so)

Harder Problem – MDVRP w/ per-vehicle weights

- Harder to setup than expected
  - Simply treating $x_{i,j}$ as the sum of some $x^{(1)}_{i,j} + x^{(2)}_{i,j} + \ldots$ just makes the LP solver choose the minimum cost at each edge
- The trick is to introduce $|D||V / D|$ binary constraints to force customer vertices to be serviced by one specific driver
- Also need extra constraints to ensure each vehicle only departs/returns to its individual depot
  - let D* := set of drivers

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Harder Problem – MDVRP w/ per-vehicle weights

$$\forall i \in V/D \quad \sum_{j \sim i} x_{i,j} = 1$$

$$\forall j \in V/D \quad \sum_{i \sim j} x_{i,j} = 1$$

$$\forall i \in V/D, k \in D^* \quad \sum_{j \sim i} x_{i,j}^{(k)} = y_i^{(k)}$$

$$\forall j \in V/D, k \in D^* \quad \sum_{i \sim j} x_{i,j}^{(k)} = y_j^{(k)}$$

$$\forall i \in V/D \quad \sum_{k \in D^*} y_i^{(k)} = 1$$

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Harder Problem – MDVRP w/ per-vehicle weights

$$\forall i \in D \quad \sum_{j \sim i} x_{i,j} = d_i$$

$$\forall j \in D \quad \sum_{i \sim j} x_{i,j} = d_i$$

$$\forall k \in D^*, i = \text{depot of } k \quad \sum_{j \sim i} x_{i,j}^{(k)} = 1$$

$$\forall k \in D^*, j = \text{depot of } k \quad \sum_{i \sim j} x_{i,j}^{(k)} = 1$$

$$\forall k \in D^* \quad \sum_{i \in D, i \neq \text{depot of } k} \sum_{j \sim i} x_{i,j}^{(k)} = 0$$

$$\forall k \in D^* \quad \sum_{j \in D, j \neq \text{depot of } k} \sum_{i \sim j} x_{i,j}^{(k)} = 0$$

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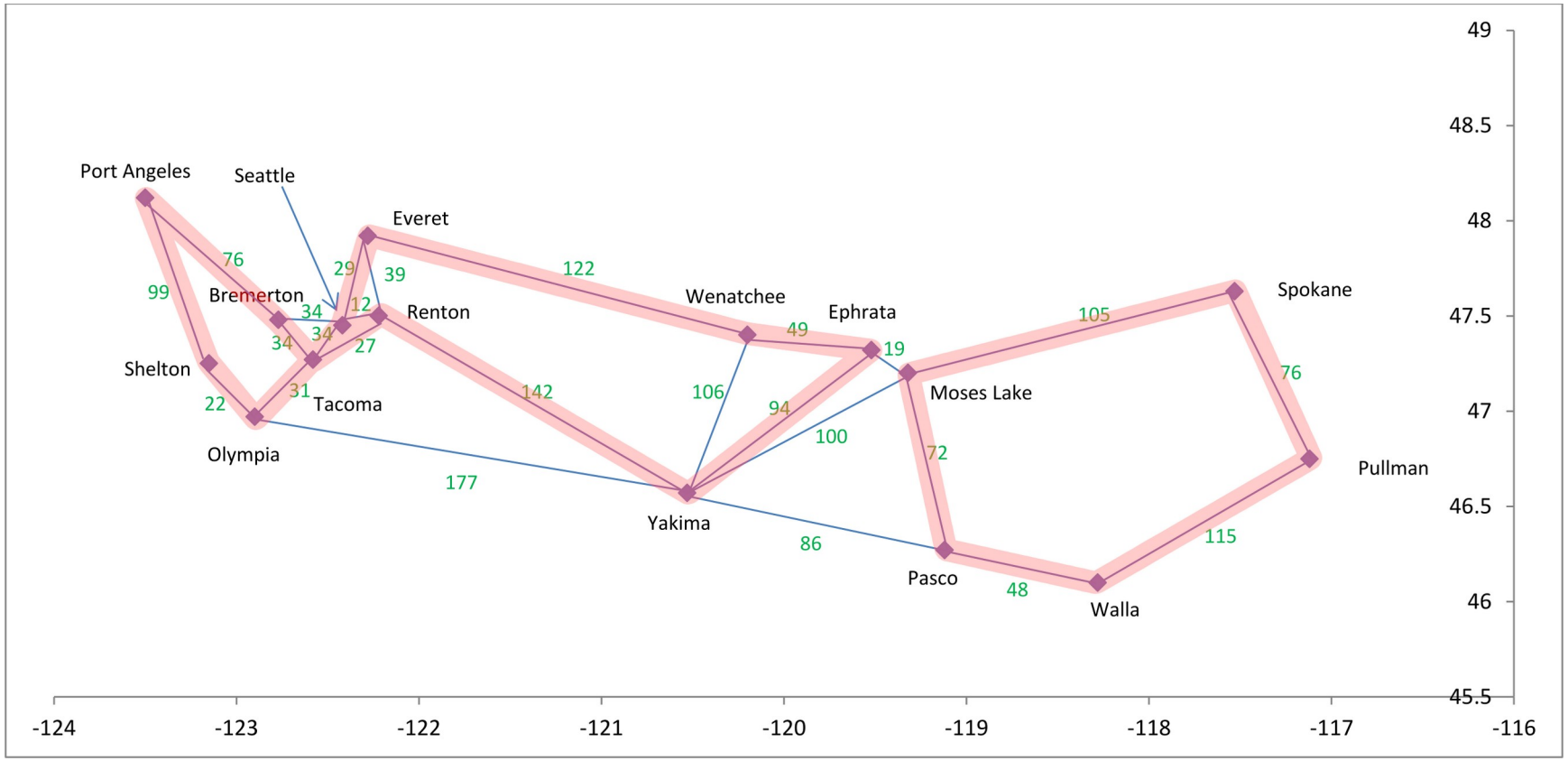
Harder Problem – MDVRP w/ per-vehicle weights



Very Hard Problem – MDVRP w/ per-vehicle weights on a real road network

- Let's construct a more "realistic" MDVRP to solve
- OpenStreetMap data for Knox County, TN [2]
  - 143336 vertices
  - 3373605 edges
  - Our new problem will have 12 customers and 3 depots
- Naively plugging this into LP solver would create $\sim 2^{143336}$ constraints
  - You would need on the order of 10^{40000} GiB just to represent in memory!

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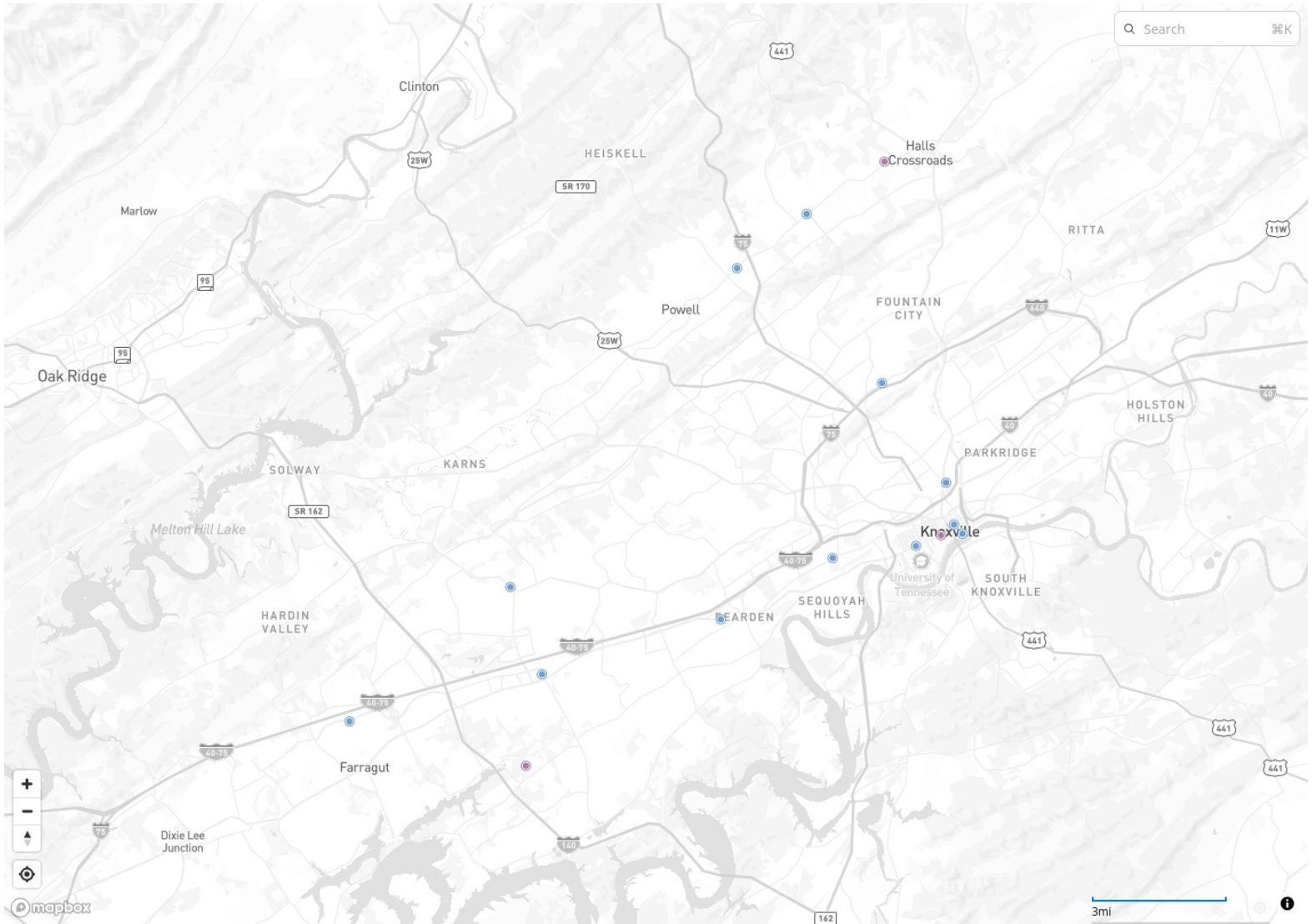
Very Hard Problem – MDVRP w/ per-vehicle weights on a real road network



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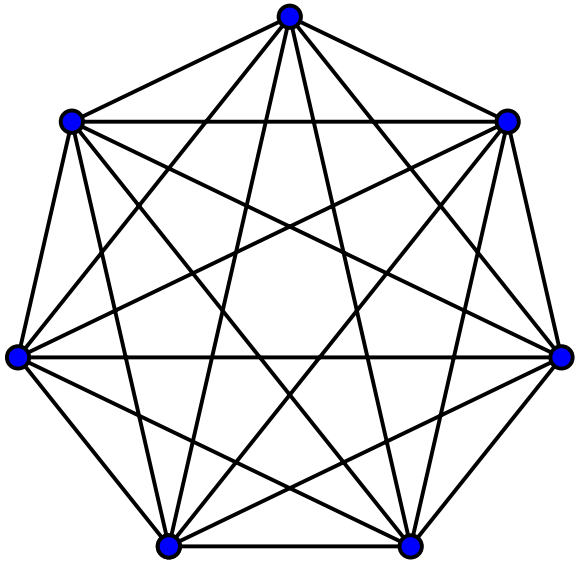
Very Hard Problem – MDVRP w/ per-vehicle weights on a real road network

- Given our set of customers/depots $U \subseteq V$, form the complete graph $K_{|U|}$ where:
  - Each vertex in $K_{|U|}$ represents a vertex in U
  - Each edge in $K_{|U|}$ represents the shortest path between two vertices in U (solvable via A*)
- Solve the mixed-integer LP for $K_{|U|}$



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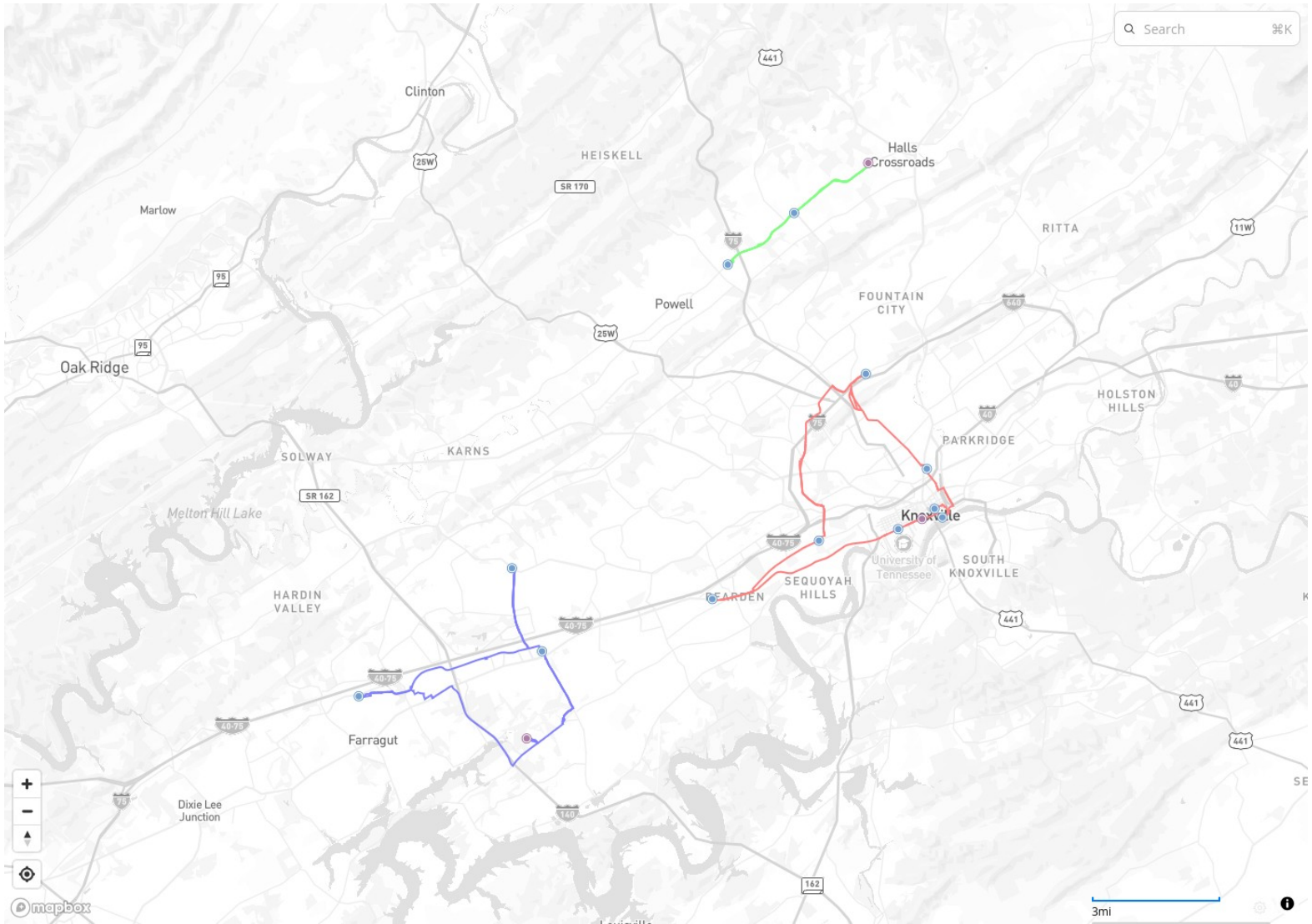
Very Hard Problem – MDVRP w/ per-vehicle weights on a real road network



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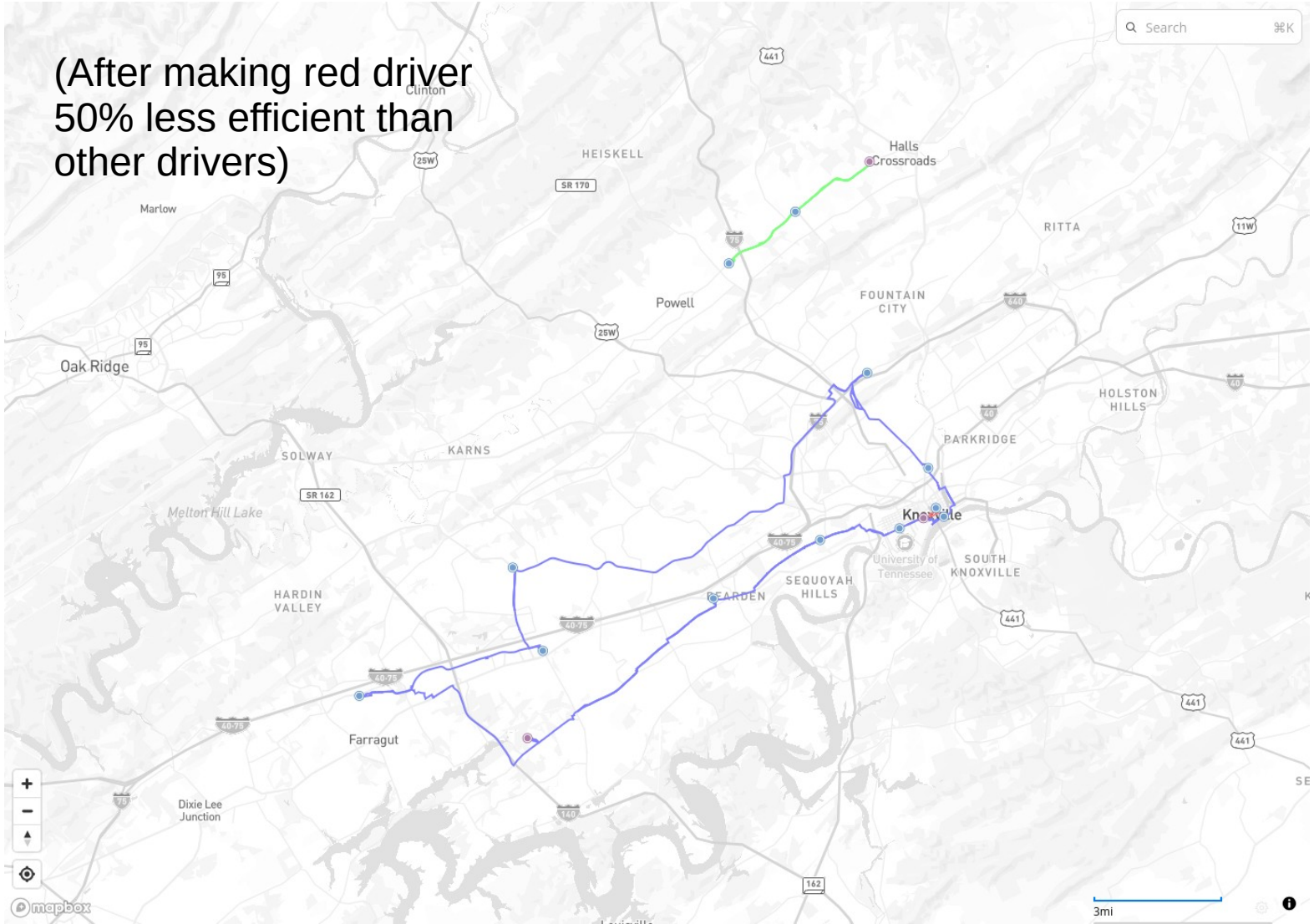
Very Hard Problem – MDVRP w/ per-vehicle weights on a real road network



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Practical Considerations

- Linear differences in weights leads to one vehicle dominating all others
  - A time-constrained Lagrangean relaxation may solve this
  - Alternatively, we can adjust the depot constraints if we don't care whether or not all vehicles visit at least one customer
- On such a large graph, even A* is slow
  - The subproblems are embarrassingly parallel – take advantage of this
  - Great-circle distance is consistent, but expensive to calculate – **use simple heuristics**!
  - "Real-world" routing software employs lots of geospatial optimizations (e.g. contraction hierarchies) to get fast results with A* [3]

Practical Considerations

- This is still $O(2^n)$ time/space complexity
  - Some recent probabilistic/heuristic-based approximations used for certain cases of the VRP:
    - Swarm intelligence and genetic algorithms [4]
    - Simulated annealing [5]
    - Adaptive Large-Neighborhood Search [6]

References

[1] P. Toth and D. Vigo, *The Vehicle Routing Problem*, in SIAM Monographs on Discrete Mathematics and Applications, Philadelphia, PA, USA: Society for Industrial and Applied Mathematics, 2002, doi:10.1137/1.9780898718515.

[2] Various OpenStreetMap contributors. “tennessee-latest.osm.pbf.” Distributed by Geofabrik GmbH Karlsruhe. https://download.geofabrik.de/north-america/us/tennessee.html (accessed Apr 23, 2026).

[3] GraphHopper GmbH, "GraphHopper," ver. 11.0, Open Source, Oct 2025, [Online]. Available: https://github.com/graphhopper/graphhopper/releases/tag/11.0.

[4] Wang, Chun Bo, and Jun Guo, “A New Hybrid Algorithm Based on Artificial Fish Swarm Algorithm and Genetic Algorithm for VRP.” *Applied Mechanics and Materials*, vol. 325–326, pp. 1722–25, Jun 2013, doi:10.4028/www.scientific.net/AMM.325-326.1722.

[5] Lin, Shih-Wei, et al,“Solving the Truck and Trailer Routing Problem Based on a Simulated Annealing Heuristic.” *Computers & Operations Research*, vol. 36, no. 5, pp. 1683–92, May 2009, doi:10.1016/j.cor.2008.04.005.

[6] R. Stefan, and D. Pisinger, “A Unified Heuristic for a Large Class of Vehicle Routing Problems with Backhauls.” *European Journal of Operational Research*, vol. 171, no. 3, pp. 750–75, Jun 2006, doi:10.1016/j.ejor.2004.09.004.

Questions/Comments?

Code: https://github.com/TheHashTableSlasher/ece8920_final